LABORATORY OBSERVATION

Course: Full Stack Web Development
Course Code: 23CM4121
Branch: CSM
Section: B
Task No.: 1
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1. TITLE

Design and Develop a College Information Portal Using HTML5 to Demonstrate Text Formatting, Lists, Tables, Forms, and Multimedia Elements

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2. AIM

To design and develop a static HTML5 web page — the "College Information Portal" — that demonstrates the use of semantic tags, text formatting elements, ordered and unordered lists, hyperlinks, a structured table, images, an interactive registration form, HTML5 input types, and embedded audio/video elements.

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3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 document.
- To use semantic tags such as section, article, and footer to organise page content.
- To apply text formatting tags such as b, i, u, sup, sub, and small.
- To create ordered and unordered lists to present information.
- To insert hyperlinks that open external websites in a new tab.
- To design a table to display structured student data.
- To embed images with meaningful alt text.
- To build an interactive registration form using text, email, password, radio, checkbox, select, and textarea inputs.
- To use HTML5 input types such as date, number, color, and range.
- To embed audio and video elements using native HTML5 support.

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4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web pages.

HTML5 is the latest version of HTML. It introduces semantic elements and new form input types that make pages more meaningful and interactive, without needing external plugins or JavaScript.

In this page, semantic and formatting tags are used to structure content and present it clearly to the browser.

- **`section, article, footer`** are used to organise related content into logical blocks.
- **`sup and sub`** raise or lower text — used for symbols, ordinal suffixes, and chemical formulas.
- **`ol and ul`** create ordered and unordered lists respectively.
- **`a href="..."`** creates a hyperlink to another page or website.
- **`table, tr, th, td`** define a table, its rows, header cells, and data cells.
- **`form, input, select, textarea`** create interactive elements for user input.
- **`audio and video`** embed native media players without external plugins.

The basic flow of the page is:

Browser requests HTML file → Browser parses HTML → Elements are rendered as a structured web page

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5. ALGORITHM / PROCEDURE

1. Create a new HTML file named index.html.
2. Add the DOCTYPE declaration and set the language attribute.
3. Add the head section with meta tags and a page title.
4. Create a section with formatted text, using bold, italic, underline, superscript, and subscript.
5. Add ordered and unordered lists for courses and facilities.
6. Insert hyperlinks to external websites, set to open in a new tab.
7. Create a table to display student roll number, name, branch, and email.
8. Insert images for the college logo and campus.
9. Build a registration form using text, email, password, radio, checkbox, select, and textarea inputs.
10. Add HTML5 input types such as date, number, color, and range.
11. Embed audio and video elements using the audio and video tags.
12. Add a footer with copyright information.
13. Save the file and open it in a web browser.
14. Verify that all elements render and function correctly.

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6. PROGRAM

`index.html`

```html
<!DOCTYPE html>
<!-- Declares this as an HTML5 document -->
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>College Information Portal</title>
</head>
```

```html
<body style="background-color: bisque;">

<!-- ================= PART A: TEXT FORMATTING, LISTS & LINKS ================= -->
<section id="about">
     <h1 style="color: #2c3e50;">ANITS Engineering College</h1><hr>
    <p>Welcome to our college portal. ANITS Engineering College<sup>&reg;</sup>
       <!-- <sup> raises text above the line — used here for the ® symbol,
            which is always written slightly higher than normal text -->
       was established in 1998 and offers <b>quality education</b>,
       <!-- <b> makes text bold, just to highlight a keyword -->
       <i>modern infrastructure</i>, and <u>excellent placements</u>.
       <!-- <i> italicizes, <u> underlines — pure text formatting,-->
       The college is currently ranked 3<sup>rd</sup> among private engineering
       <!-- <sup> again — this time for the ordinal suffix "rd" in "3rd",
            which is always written smaller and raised in real writing -->
       colleges in the state.
    </p>

    <h2 id="courses" style="color: #2c3e50;">Courses Offered</h2>
    <!-- Ordered list: sequence/ranking matters here (as displayed on the
         admission brochure, courses are listed in order of intake priority) -->
    <ol>
        <!-- <ol> = Ordered List. Browser numbers items automatically (1,2,3...) -->
        <li>Computer Science Engineering</li>
        <li>Electronics Engineering</li>
        <li>Mechanical Engineering</li>
        <li>Chemical Engineering</li>
    </ol>

    <h2 style="color: #2c3e50;">Facilities</h2>
    <!-- Unordered list: no ranking implied, just a set of facilities -->
    <ul>
        <!-- <ul> = Unordered List. Browser just adds bullet points, no numbering -->
        <li>Central Library</li>
        <li>Sports Complex</li>
        <li>Hostel</li>
        <li>Chemistry Laboratory</li>
    </ul>

    <p>In the Chemistry Laboratory, Chemical Engineering students carry out
       experiments involving compounds such as H<sub>2</sub>SO<sub>4</sub>
       <!-- <sub> lowers text below the line — needed here because chemical
            formula numbers are always written smaller and below the letter -->
       (sulphuric acid titrations) and study the behaviour of gases such as
       CO<sub>2</sub>, as part of their core lab curriculum.</p>

    <p><small><sup>1</sup>As per the State Engineering Colleges Ranking
       <!-- <small> makes footnote-style text smaller, and <sup> marks the
            "1" as a footnote reference number, same idea as in a printed book -->
```

```html
        Survey, 2025.</small></p>

    <p>
        <a href="https://www.college-example.edu" target="_blank">Visit College Website</a> |
        <!-- <a href="..."> creates a clickable link.
             target="_blank" makes it open in a NEW browser tab
             instead of navigating away from the current page -->
        <a href="https://www.google.com" target="_blank">Google</a>
    </p>
</section>

<!-- ================= PART B: TABLES AND IMAGES ================= -->
<section id="students">
    <h2 style="color: #2c3e50;">Student Details</h2>
    <table border="1" cellpadding="8">
        <!-- <table> holds tabular data. border and cellpadding are only
             for visibility while learning — real projects use CSS instead -->
        <tr>
            <!-- <tr> = Table Row -->
            <th>Roll No</th>
            <!-- <th> = Table Header cell — bold and centered by default,
                 used only in the first row to label columns -->
            <th>Name</th>
            <th>Branch</th>
            <th>Email</th>
        </tr>
        <tr>
            <td>101</td>
            <!-- <td> = Table Data cell — holds the actual values -->
            <td>Niharika</td>
            <td>CSE</td>
            <td>niharika@example.com</td>
        </tr>
        <tr>
            <td>102</td>
            <td>Saranya</td>
            <td>ECE</td>
            <td>sara@example.com</td>
        </tr>
        <tr>
            <td>103</td>
            <td>Divya</td>
            <td>Chemical</td>
            <td>divya@example.com</td>
        </tr>
    </table>

    <h2 style="color: #2c3e50;">College Logo &amp; Campus</h2>
    <img src="logo.jpg" alt="ANITS Engineering College Logo" width="120">
```

```html
    <!-- alt is NOT just decoration — it's what shows up if the image
         fails to load, and what a screen reader speaks aloud for
         visually impaired users. It should always describe the image. -->
    <img src="campus.jpg" alt="Aerial view of ANITS Engineering College campus" width="300">
</section>

<!-- ================= PART C: STUDENT REGISTRATION FORM ================= -->
<section id="register">
    <h2 style="color: #2c3e50;">Student Registration Form</h2>
    <form>
        <label>Name:</label>
        <input type="text" name="sname" required><br><br>
        <!-- "required" is HTML5 built-in validation — the browser itself
             blocks submission if this field is left empty, no JS needed -->

        <label>Email:</label>
        <input type="email" name="semail" required><br><br>
        <!-- type="email" checks that the text looks like a valid email
             (has an @ and a domain) before allowing submission -->

        <label>Password:</label>
        <input type="password" name="spass" required><br><br>
        <!-- type="password" hides the typed characters as dots -->

        <label>Gender:</label>
        <input type="radio" name="gender" value="Male"> Male
        <input type="radio" name="gender" value="Female"> Female<br><br>
        <!-- Both radios share name="gender" — that's what links them
             together as ONE group, so only one can ever be selected -->

        <label>Hobbies:</label>
        <input type="checkbox" name="hobby" value="Reading"> Reading
        <input type="checkbox" name="hobby" value="Sports"> Sports<br><br>
        <!-- Checkboxes work independently of each other, so ticking one
             does NOT untick the other — multiple selections are allowed -->

        <label>Branch:</label>
        <select name="branch">
            <!-- <select> + <option> = a dropdown menu, saves screen space
                 compared to listing all choices as radio buttons -->
            <option value="CSE">CSE</option>
            <option value="ECE">ECE</option>
            <option value="MECH">MECH</option>
            <option value="CHEM">Chemical</option>
        </select><br><br>

        <label>Comments:</label><br>
        <textarea rows="4" cols="30"></textarea><br><br>
        <!-- <textarea> allows multi-line free text, unlike <input type="text">
```

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            which only allows a single line -->

        <input type="submit" value="Submit">
        <!-- Sends all form data (to a server, in a real app) -->
        <input type="reset" value="Reset">
        <!-- Clears every field back to its original blank/default state -->
    </form>
</section>

<!-- ================= PART D: HTML5 FEATURES ================= -->
<section id="notice">
    <!-- <section> groups related content together as one thematic block -->
    <article>
        <!-- <article> = content that could stand alone and still make sense
             on its own, like a notice or a news post -->
        <h2 style="color: #2c3e50;"><i>Latest Notice</i></h2>
        <p>Semester exams start from next month. Chemical Engineering students
           should collect their lab coats before the practical session.</p>
    </article>

    <h2 style="color: #2c3e50;">Media</h2>
    <audio controls src="welcome.mp3"></audio><br>
    <!-- controls = shows the built-in play/pause/volume bar automatically,
         no custom player needed -->
    <video controls width="300" src="campus_tour.mp4"></video>

    <h2 style="color: #2c3e50;">HTML5 Input Types</h2>
    <form>
        Email: <input type="email"><br>
        Date of Birth: <input type="date"><br>
        <!-- type="date" gives a calendar picker instead of a plain text box -->
        Age: <input type="number" min="17" max="30"><br>
        <!-- type="number" restricts input to numbers and can enforce a range -->
        Favourite Color: <input type="color"><br>
        <!-- type="color" opens a color-picker swatch -->
        Rate our campus tour: <input type="range" min="1" max="5">
        <!-- type="range" gives a slider instead of typing a number -->
    </form>
</section>
<footer>
    <!-- <footer> marks the closing section of the page, e.g. copyright info -->
    <p>&copy; 2026 ANITS Engineering College</p>
</footer>
</body>
</html>
```

7. CODE EXPLANATION

Code	Explanation
<section>, <article>, <footer>	Group related content into semantic, meaningful blocks.
<sup>, <sub>	Raise or lower text for symbols, ordinal suffixes, and chemical formulas.
<b>, <i>, <u>	Bold, italicise, and underline text for visual emphasis.
, ,	Create ordered and unordered lists and their items.
<a href="..." target="_blank">	Creates a hyperlink that opens in a new browser tab.
<table>, <tr>, <th>, <td>	Define a table, its rows, header cells, and data cells.
<img src="" alt="">	Embeds an image with alternate text for accessibility.
<form>, <input>, <select>, <textarea>	Create the interactive elements of the registration form.
required	HTML5 built-in validation that blocks submission if a field is empty.
type="radio" / "checkbox"	Radio buttons allow one choice per group; checkboxes allow multiple.
type="date" / "number" / "color" / "range"	HTML5 input types for specialised data entry.
<audio controls>, <video controls>	Embed native audio/video players without external plugins.

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8. TEST CASES AND EXECUTION DATA

The page was opened in a browser and each interactive element was tested individually; the results were recorded below.

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open index.html in a browser	Page loads and displays the college portal	Page displayed correctly	Pass

TC-02	Leave the Name field empty and click Submit	Browser blocks submission and prompts for the field	Required-field prompt shown	Pass
TC-03	Enter an invalid email format	Browser shows an email-format validation error	Validation error displayed	Pass
TC-04	Select both Reading and Sports checkboxes	Both remain selected at the same time	Both checkboxes selected	Pass
TC-05	Click 'Visit College Website' link	Link opens the website in a new browser tab	New tab opened correctly	Pass
TC-06	Drag 'Rate our campus tour' slider	Slider value updates as it is dragged	Slider value updated	Pass

Additional Validation Test

Test Case	Input/Action	Expected Result	Status
TC-07	Open the Date of Birth field	Browser shows a calendar date-picker	Pass
TC-08	Open the Favourite Color field	Browser shows a colour-picker swatch	Pass

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9. OUTPUT

Output1:Basic HTML

ANITS Engineering College

Welcome to our college portal. ANITS Engineering College® was established in 1998 and offers **quality education**, *modern infrastructure*, and excellent placements. The college is currently ranked 3rd among private engineering colleges in the state.

Courses Offered

1. Computer Science Engineering
2. Electronics Engineering
3. Mechanical Engineering
4. Chemical Engineering

Facilities

- Central Library
- Sports Complex
- Hostel
- Chemistry Laboratory

In the Chemistry Laboratory, Chemical Engineering students carry out experiments involving compounds such as H_2SO_4 (sulphuric acid titrations) and study the behaviour of gases such as CO_2, as part of their core lab curriculum.

[1]As per the State Engineering Colleges Ranking Survey, 2025.

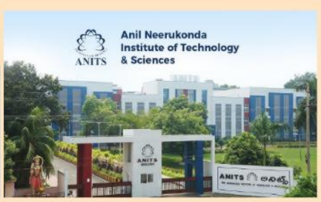
Visit College Website | Google

Output2:Table & images

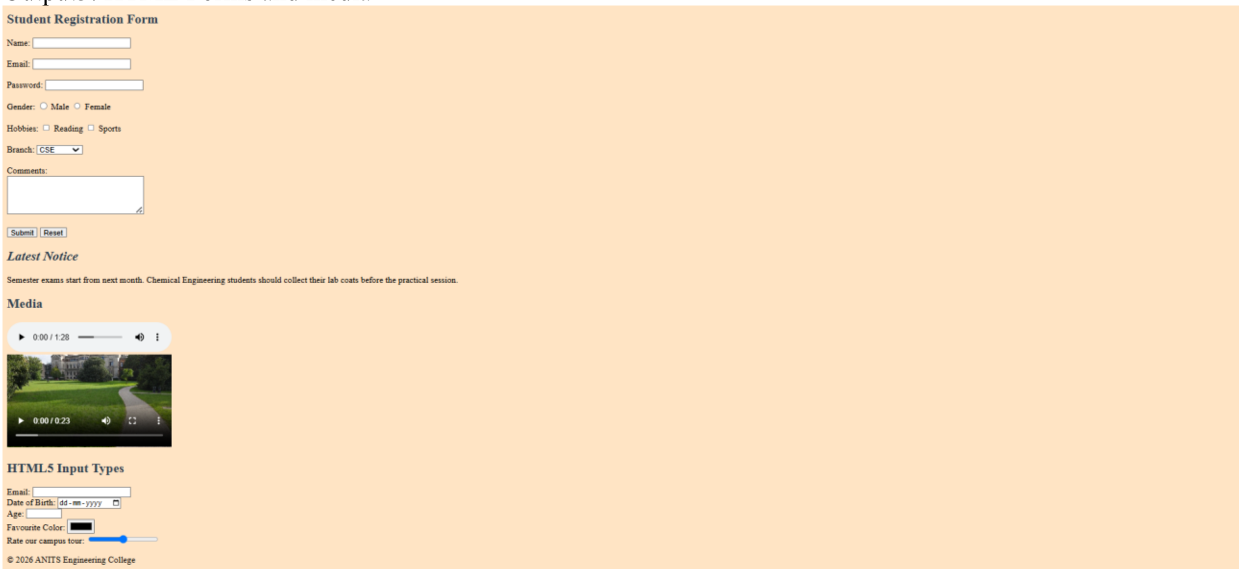
Student Details

Roll No	Name	Branch	Email
101	Niharika	CSE	niharika@example.com
102	Saranya	ECE	sara@example.com
103	Divya	Chemical	divya@example.com

College Logo & Campus



Output3: HTML Forms and media



10. RESULT

Thus, a College Information Portal was successfully designed and developed using HTML5. Semantic tags, text formatting, ordered and unordered lists, hyperlinks, a structured table, images, a registration form, HTML5 input types, and multimedia elements were implemented and verified to render and function correctly in the web browser.